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Reference

video

OpenCourseWare - Mathematics for Computer Science - 2. induction

Lecture hand written Notes

Lecture 2: induction

direct proofs - you drive it. is true by axioms and logis

Proof by contradiction

To prove P is True, we assume P is False (i.e ¬P is T) & then use that hypothesis to derive a falsehood or contrdiction.

If $\neg P \Rightarrow F$ is true

$\neg P$

↓

F

↓

P = True

Ex Thm: $\sqrt{2}$ is irrational

Pf (by Cont.) ← need to write that down always. assume for P = F

Assume fro purpose of contradiction that $\sqrt{2}$ is rational ↵

$\sqrt{2} = \frac{a}{b}$ (fraction in lowest terms)

$2 = \frac{a^2}{b^2}$

$2b^2 = a^2$

$\Rightarrow a$ is even ($2 \mid a$) ← divide symbol

↘ b.c $a^2 = \underline{2}b^2$

$\Rightarrow 4 \mid 2b^2$

$\Rightarrow 2 \mid b^2$

$\Rightarrow b$ is even

$\Rightarrow a/b$ is not in lowest terms

$\Rightarrow$ contradiction (*)

$\Rightarrow \sqrt{2}$ is irrelational. $\square$

Induction axiom

Let $P(n)$ be a predicate. If $P(0) = T$ and $\forall n \in \mathbb{N}\ (P(n) \Rightarrow P(n+1)) = T$

then $\forall n \in \mathbb{N}\ P(n) = T$

If $P(0), P(0) \Rightarrow P(1), P(1) \Rightarrow P(2), \ldots$ are true, then

$P(0), P(1), P(2) \ldots$ are true

like an series of domino

Thm: $\forall n \geq 0 \quad 1+2+3+\cdots+n = \frac{n(n+1)}{2}$

$$\sum_{i=1}^{n} i = \sum_{i \leq 1 \leq n} 1 = \sum_{i \leq 1 \leq n} i$$

If $n=1 \Rightarrow 1$

If $n \leq 0 \Rightarrow 0$

$n=4$, $1+2+3+4 = 10 = \frac{4(5)}{2}$

Pf: by induction ← need to write it ✓ state your induction hypothesis

let $P(n)$ be proposition that $\sum_{i=1}^{n} i = \frac{n(n+1)}{2}$

1. check the base case $P(0)$

Base Case: $P(0)$ is true

$$\sum_{i=1}^{0} i = 0 = \frac{0(0+1)}{2}$$

inductive step: for $n \geq 0$, show $P(n) \Rightarrow P(n+1)$ is true

Assume $P(n)$ is true for purpose of induction

(ie, assume $1+2+\cdots+n = \frac{n(n+1)}{2}$)

$\frac{(n+1)(n+2)}{2}$

need to show $1+2+\cdots+(n+1) = \frac{\cdots}{2}$

$$\underbrace{1+2+\cdots+n}_{\frac{n(n+1)}{2}}+(n+1)$$

$$\frac{n(n+1)}{2}+n+1 = \frac{n^2+n+2n+2}{2} = \frac{(n+1)(n+2)}{2}$$ ✓

$\forall n \geq 0$

$P(n) \Rightarrow P(n+1)$

~~#~~ induction help us prove the theorem but it may not help us understand why the theorem is true

Thm: $\forall n \in \mathbb{N}$ $3 \mid (n^3 - n)$ Ex $n = 5$ $3 \mid (125 - 5)$ $\frac{n^3 - n}{3}$

pf: by induction

let $P(n)$ $3 \mid (n^3 - n)$

Base Case $n = 0$ $3 \mid (0 - 0)$

Inductive step For $n \geq 0$, show $P(n) \Rightarrow P(n+1)$ is T

Assume $P(n)$ is T, i.e $3 \mid (n^3 - n)$

Examine $(n+1)^3 - (n+1) = n^3 + 3n^2 + 3n + 1 - (n+1)$

* you need to use $3 \mid (n^3 - n)$ to prove $P(n+1)$ is T

$= n^3 + 3n^2 + 2n$

$= n^3 - n + 3n^2 + 3n$

$3 \mid 3n^2$

$3 \mid 3n$

$3 \mid n^3 - n$ by Inductive Hypothesis $P(n)$

$3 \mid (n+1)^3 - (n+1)$

$\forall n \geq 0$, $P(n) \Rightarrow P(n+1)$

need to follow up

you can start every where not just zero but go in ind step

Base Case : P(b) is True

Ind step : $\forall n \geq b$ $P(n) \Rightarrow P(n+1)$

Conclude $\forall n \geq b$ $P(n)$

Thm (not!) All horses are the same color

↖ we can't use it for predicate b.c you can't plug any value

Pf: by induction.

P(n) = In any set of n horses, the horses are all the same color

Base Case

P(1) true since just 1 horse

Inductive step

Assume P(n) is T to prove P(n+1)

Consider a set of n+1 hourse, $H_1, H_2, \ldots, H_{n+1}$

Then $H_1, H_2, \ldots, H_n$ are the same color (since P(n)=T assume)

& $H_2, H_3, \ldots, H_{n+1}$ " "

Since color (H_1) = color $(H_2, \ldots, H_n)$ = color (H_{n+1})

⇒ n+1 are the same color ⇒ P(n+1)

↗

Flow in horse proof

you need to establish the inductive step for all $n \geq$ base case

P(1) and P(2)

- there is no link bet

P(1), } P(2) ⇒ P(3), ...

$\forall n \geq 2$ P(n) ⇒ P(n+1)

P(1) ⇒ P(2)

there is no proof for that

⚠ always check the base case

Tilling proof by induction

how induction can be used to prove there is a solution to a problem
and also how to find a solution

Thm $\forall n$, $\exists$ way to tile a $2^n \times 2^n$ region with a ~~center~~ any □ square missing (for Bill)

Pf: by induction

$P(n)$

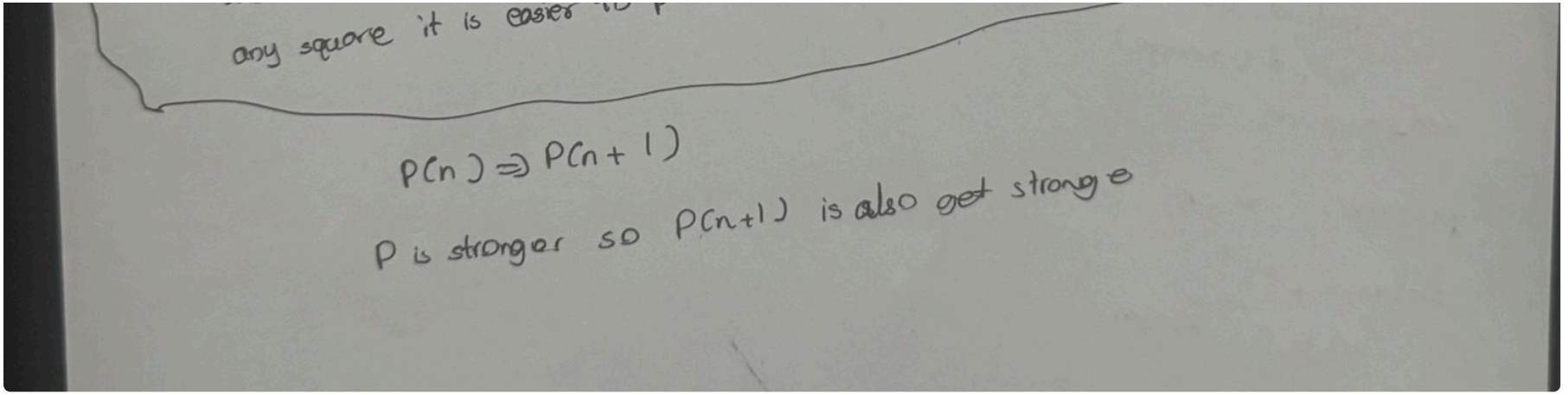
Base Case $P(0) = \square$ ✓

inductive step for $n \geq 0$, assume $P(n)$ to verify I.H

so we need to show $P(n+1)$ is true

Consider a $2^{n+1} \times 2^{n+1}$ courtyard

if you are struggling what's the induction hypothesis to use.
If what you've got doesn't work, you could pick a differenet one,
but better to pick a stronger one
we have trouble to prove with center, coner square, by making it stronger,
to prove



Reading : 3. Induction

These methods are especially useful when you need to prove that a predicate is true for all natural numbers.

- Well Ordering Principle,
- the Induction Rule, and
- Strong Induction.

even though these methods seem different, but they are equivalent that if you prove using one of the ways, you reform it to other two ways.

The choice of which method to use is up to you and typically depends on whichever seems to be the easiest or most natural for the problem at hand.

3.1 The Well Ordering Principle

Every nonempty set of nonnegative integers has a smallest element.

it have to be *nonempty* and *non-negative integers*

The clever idea in this proof is: if something impossible exists, choose the smallest example of it, then show that an even smaller example exists. That contradiction proves no such example exists.

3.1.1 Well Ordering Proofs

every fraction can be written in lowest terms

proof by Contradion

Assume for the purpose of contradition that

at least one can't be written in lowest terms

if numerator and denominator had no common factor greater than 1, than that function is in lowest terms

since it can't be written in lowest term, $\gcd(m_0, n_0) > 1$

we will define set C:

$C = \{$ postive integer that are numerators of bad functions$\}$

we will choose smallest element of C since WOP guaratees that it have smallest element. call it m_0

$m_0 \in C \Rightarrow$ we also has $n_0 \in C, n_0 > 0$

$\frac{m_0}{n_0}$

we can divid both m_0 and n_0 by $\gcd(m_0, n_0)$

$\frac{m_0/p}{n_0/p}$ can't be written in lowest form either

m_0/p and ~~n_0~~ n_0/p is lowest which contrdicts to m_0

So C must be empty set $\Rightarrow$ no numerators of fraction that can't be written in lowest terms $\Rightarrow$ no function at all ∎

3.1.2 Template for Well Ordering Proofs

3.1.2 Template for Well Ordering Proofs

More generally, to prove that "$P(n)$ is true for all $n \in \mathbb{N}$" using the Well Ordering Principle, you can take the following steps:

- Define the set, C, of *counterexamples* to P being true. Namely, define[2]

$$C ::= \{n \in \mathbb{N} \mid P(n) \text{ is false}\}.$$

- Use a proof by contradiction and assume that C is nonempty.
- By the Well Ordering Principle, there will be a smallest element, n, in C.
- Reach a contradiction (somehow)—often by showing how to use n to find another member of C that is smaller than n. (This is the open-ended part of the proof task.)
- Conclude that C must be empty, that is, no counterexamples exist. QED

3.1.3 Examples

Theorem 3.1.2. *Every natural number can be factored as a product of primes.*

Proof. By contradiction and Well Ordering. Assume that the theorem is false and let C be the set of all integers greater than one that cannot be factored as a product of primes. We assume that C is not empty and derive a contradiction.

If C is not empty, there is a least element, $n \in C$, by Well Ordering. The n can't be prime, because a prime by itself is considered a (length one) product of primes and no such products are in C.

So n must be a product of two integers a and b where $1 < a, b < n$. Since a and b are smaller than the smallest element in C, we know that $a, b \notin C$. In other words, a can be written as a product of primes $p_1 p_2 \cdots p_k$ and b as a product of primes $q_1 \cdots q_l$. Therefore, $n = p_1 \cdots p_k q_1 \cdots q_l$ can be written as a product of primes, contradicting the claim that $n \in C$. Our assumption that C is not empty must therefore be false. ■

3.2 Ordinary Induction

Induction is by far the most powerful and commonly-used proof technique in discrete mathematics and computer science.

Ordinary induction explain

let $P(n)$ be a predicate. If $P(0) = T$ and $\forall n \in \mathbb{N}(P(n) \implies P(n+1)) = T$ then $\forall n \in \mathbb{N} = T$

3.2.1 A Rule for Ordinary Induction

The Principle of Induction.

Let $P(n)$ be a predicate. If

- $P(0)$ is true, and
- $P(n)$ IMPLIES $P(n+1)$ for all nonnegative integers, n,

then

- $P(m)$ is true for all nonnegative integers, m.

Rule. ***Induction Rule***

$$\frac{P(0), \quad \forall n \in \mathbb{N}.\, P(n) \text{ IMPLIES } P(n+1)}{\forall m \in \mathbb{N}.\, P(m)}$$

This general induction rule works for the same intuitive reason that all the students get candy bars, and we hope the explanation using candy bars makes it clear why the soundness of the ordinary induction can be taken for granted. In fact, the rule is so obvious that it's hard to see what more basic principle could be used to justify it.[3] What's not so obvious is how much mileage we get by using it.

3.2.3 A Template for Induction Proofs

The proof of Theorem 3.2.1 was relatively simple, but even the most complicated induction proof follows exactly the same template. There are five components:

1. **State that the proof uses induction.** This immediately conveys the overall structure of the proof, which helps the reader understand your argument.

2. **Define an appropriate predicate $P(n)$.** The eventual conclusion of the induction argument will be that $P(n)$ is true for all nonnegative n. Thus, you should define the predicate $P(n)$ so that your theorem is equivalent to (or follows from) this conclusion. Often the predicate can be lifted straight from the proposition that you are trying to prove, as in the example above. The predicate $P(n)$ is called the *induction hypothesis*. Sometimes the induction hypothesis will involve several variables, in which case you should indicate which variable serves as n.

3. **Prove that $P(0)$ is true.** This is usually easy, as in the example above. This part of the proof is called the *base case* or *basis step*.

4. **Prove that $P(n)$ implies $P(n+1)$ for every nonnegative integer n.** This is called the *inductive step*. The basic plan is always the same: assume that $P(n)$ is true and then use this assumption to prove that $P(n+1)$ is true. These two statements should be fairly similar, but bridging the gap may require some ingenuity. Whatever argument you give must be valid for every nonnegative integer n, since the goal is to prove the implications $P(0) \to P(1)$, $P(1) \to P(2)$, $P(2) \to P(3)$, etc. all at once.

5. **Invoke induction.** Given these facts, the induction principle allows you to conclude that $P(n)$ is true for all nonnegative n. This is the logical capstone to the whole argument, but it is so standard that it's usual not to mention it explicitly.

Always be sure to explicitly label the *base case* and the *inductive step*. It will make your proofs clearer, and it will decrease the chance that you forget a key step (such as checking the base case).

3.2.4 A Clean Writeup

3.2.4 A Clean Writeup

The proof of Theorem 3.2.1 given above is perfectly valid; however, it contains a lot of extraneous explanation that you won't usually see in induction proofs. The writeup below is closer to what you might see in print and should be prepared to produce yourself.

Proof of Theorem 3.2.1. We use induction. The induction hypothesis, $P(n)$, will be equation (3.2).

Base case: $P(0)$ is true, because both sides of equation (3.2) equal zero when $n = 0$.

Inductive step: Assume that $P(n)$ is true, where n is any nonnegative integer. Then

$$\begin{aligned} 1 + 2 + 3 + \cdots + n + (n+1) &= \frac{n(n+1)}{2} + (n+1) && \text{(by induction hypothesis)} \\ &= \frac{(n+1)(n+2)}{2} && \text{(by simple algebra)} \end{aligned}$$

which proves $P(n+1)$.

So it follows by induction that $P(n)$ is true for all nonnegative n. ■

> for induction : "If you can't prove something, try to prove something grander!"

3.2.5 A More Challenging Example

Proof (successful attempt). The proof is by induction. Let $P(n)$ be the proposition that for every location of Bill in a $2^n \times 2^n$ courtyard, there exists a tiling of the remainder.

Base case: $P(0)$ is true because Bill fills the whole courtyard.

Inductive step: Assume that $P(n)$ is true for some $n \geq 0$; that is, for every location of Bill in a $2^n \times 2^n$ courtyard, there exists a tiling of the remainder. Divide the $2^{n+1} \times 2^{n+1}$ courtyard into four quadrants, each $2^n \times 2^n$. One quadrant contains Bill (**B** in the diagram below). Place a temporary Bill (**X** in the diagram) in each of the three central squares lying outside this quadrant as shown in Figure 3.4.

Now we can tile each of the four quadrants by the induction assumption. Replacing the three temporary Bills with a single L-shaped tile completes the job. This proves that $P(n)$ implies $P(n+1)$ for all $n \geq 0$. Thus $P(m)$ is true for all $n \in \mathbb{N}$, and the theorem follows as a special case where we put Bill in a central square. ■

This proof has two nice properties. First, not only does the argument guarantee that a tiling exists, but also it gives an algorithm for finding such a tiling. Second, we have a stronger result: if Bill wanted a statue on the edge of the courtyard, away from the pigeons, we could accommodate him!

Strengthening the induction hypothesis is often a good move when an induction proof won't go through. But keep in mind that the stronger assertion must actually

> Strengthening the induction hypothesis is often a good move when an induction proof won't go through.
> But keep in mind that the stronger assertion must actually be true; otherwise, there isn't much hope of constructing a valid proof!

3.2.6 A Faulty Induction Proof

recall for "all horse are the same color proof" : what error did we made?

False Theorem 3.2.3. *In every set of $n \geq 1$ horses, all the horses are the same color.*

This a statement about all integers $n \geq 1$ rather ≥ 0, so it's natural to use a slight variation on induction: prove $P(1)$ in the base case and then prove that $P(n)$ implies $P(n+1)$ for all $n \geq 1$ in the inductive step. This is a perfectly valid variant of induction and is *not* the problem with the proof below.

Bogus proof. The proof is by induction on n. The induction hypothesis, $P(n)$, will be

In every set of n horses, all are the same color. (3.4)

Base case: $(n = 1)$. $P(1)$ is true, because in a set of horses of size 1, there's only one horse, and this horse is definitely the same color as itself.

Inductive step: Assume that $P(n)$ is true for some $n \geq 1$. That is, assume that in every set of n horses, all are the same color. Now consider a set of $n + 1$ horses:

$$h_1,\ h_2,\ \ldots,\ h_n,\ h_{n+1}$$

By our assumption, the first n horses are the same color:

$$\underbrace{h_1,\ h_2,\ \ldots,\ h_n,}_{\text{same color}}\ h_{n+1}$$

Also by our assumption, the last n horses are the same color:

$$h_1,\ \underbrace{h_2,\ \ldots,\ h_n,\ h_{n+1}}_{\text{same color}}$$

So h_1 is the same color as the remaining horses besides h_{n+1}—that is, $h_2, \ldots, h_n$)—and likewise h_{n+1} is the same color as the remaining horses besides h_1–h_2, $\ldots, h_n$. Since h_1 and h_{n+1} are the same color as $h_2, \ldots, h_n$, horses $h_1, h_2, \ldots, h_{n+1}$ must all be the same color, and so $P(n + 1)$ is true. Thus, $P(n)$ implies $P(n + 1)$.

By the principle of induction, $P(n)$ is true for all $n \geq 1$. ■

The first error in this argument is in the sentence that begins "So h_1 is the same color as the remaining horses besides h_{n+1}—$h_2, \ldots, h_n$)..."

The "..." notation in the expression "$h_1, h_2, \ldots, h_n, h_{n+1}$" creates the impression that there are some remaining horses (namely $h_2, \ldots, h_n$) besides h_1 and h_{n+1}. However, this is not true when $n = 1$. In that case, $h_1, h_2, \ldots, h_n, h_{n+1} = h_1, h_2$ and there are no remaining horses besides h_1 and h_{n+1}.

So h_1 and h_2 need not be the same color!

This mistake knocks a critical link out of our induction argument. We proved $P(1)$ and we *correctly* proved $P(2) \longrightarrow P(3)$, $P(3) \longrightarrow P(4)$, etc. But we failed to prove $P(1) \longrightarrow P(2)$, and so everything falls apart: we can not conclude that $P(2), P(3)$, etc., are true. And, of course, these propositions are all false; there are sets of n non-uniformly-colored horses for all $n \geq 2$.

Students sometimes claim that the mistake in the proof is because $P(n)$ is false for $n \geq 2$, and the proof assumes something false, namely, $P(n)$, in order to prove

> proof you used in induction step are also work for base case -> base case +1

3.2.7 Induction versus Well Ordering

The well-ordering principle and mathematical induction are used to prove properties, formulas, and statements about **natural numbers** (positive $integers(\mathbb{Z}^+)$ or non-negative integers $(\mathbb{N})$)

these two are so closely related that, you can reformat Well ordering into induction and vice visa.

- sometimes induction proofs are clearer because they resemble recursive procedures that reduce handling an input of size n_1 to handling one of size n.
- On the other hand, Well Ordering proofs sometimes seem more natural, and also come out slightly shorter.

> The choice of method is really a matter of style and is up to you.

Notes for Recitation 2 : Induction

Principle of Induction. Let $P(n)$ be a predicate. If

- $P(0)$ is true, and
- for all $n \in \mathbb{N}$, $P(n)$ implies $P(n+1)$,

then $P(n)$ is true for all $n \in \mathbb{N}$.

Remember that an induction proof has five parts, though the last one is often omitted:

1. Say that the proof is by induction.
2. Define the induction hypothesis, e.g. a predicate P defined on the natural numbers.
3. Handle the base case: prove that P(0) is true.
4. Handle the inductive step: prove that $P(n)$ implies $P(n+1)$ for all integers n ≥ 0.
5. Conclude that $P(n)$ is true for all $n \in N$ by the principle of induction.

We noted in Lecture that while the base case is usually n = 0, it could be any nonnegative integer, k, in which case the conclusion would simply be that $P(n)$ holds for all $n \geq k$.

Proof. We use induction. Let $P(n)$ be the proposition that:

$$1 \cdot 2 + 2 \cdot 3 + 3 \cdot 4 + \ldots + n(n+1) = \frac{n(n+1)(n+2)}{3} \tag{1}$$

Base case $n = 1$*:* $P(1)$ is true, because the left-hand side of (1) is $1 \cdot 2 = 2$, and the right-hand side is $(1 \cdot 2 \cdot 3)/3 = 2$.

Inductive step: We must show that $P(n)$ implies $P(n+1)$ for all $n \geq 1$. So assume that $P(n)$ is true, where n denotes a positive integer. Then we can reason as follows:

$$\begin{aligned}
&1 \cdot 2 + 2 \cdot 3 + \cdots + (n+1)(n+2) \\
&= [1 \cdot 2 + 2 \cdot 3 + \cdots + n(n+1)] + (n+1)(n+2) \\
&= \frac{n(n+1)(n+2)}{3} + (n+1)(n+2) && \text{by ind. hypothesis (1)} \\
&= \frac{n(n+1)(n+2) + 3(n+1)(n+2)}{3} \\
&= \frac{(n+1)(n+2)(n+3)}{3}
\end{aligned}$$

This shows that $P(n+1)$ is true, and so $P(n)$ implies $P(n+1)$ for all $n \geq 1$.

By the induction principle, $P(n)$ is true for all $n \geq 1$, which proves the claim. □